EAST WEST UNIVERSITY

Physics Lab

Department of MPS

Course Name : Engineering Physics - I

Course Code : PHY 109 LAB

Section No : 08

Group No : 05

Experiment No : 03

Name of the Experiment : To determine the value of g, the acceleration due to gravity by means of a compound pendulum.

Date of allocation : 19.11.24

Date of submission: 26.11.24

Submitted To : Md. Hedayet Ullah

Student's ID : 2024-1-60-143

Student's Name : Rakib Al Hasan

26.11.24

PHY-109 LAB

Experiment No. 03

Name of the Experiment: To determine the value of g, the acceleration due to gravity by means of a compound pendulum.

Theory :

Compound pendulum is a rigid body of any shape free to ~~turn~~ turn about a horizontal axis. In the figure-1, Let O be the point of suspension of the body (Compound pendulum) through which passes a horizontal ~~oscillates~~ axis, perpendicular to the ~~paint~~ plane of paper, about which the body oscillates. Point G is the center of gravity of the body. Let the body be displaced through an angle θ. Then the couple acting on the body due to its weight Mg will obviously be $Mgl\sin\theta$, tending to bring back ~~into~~ its original position (vertical position).

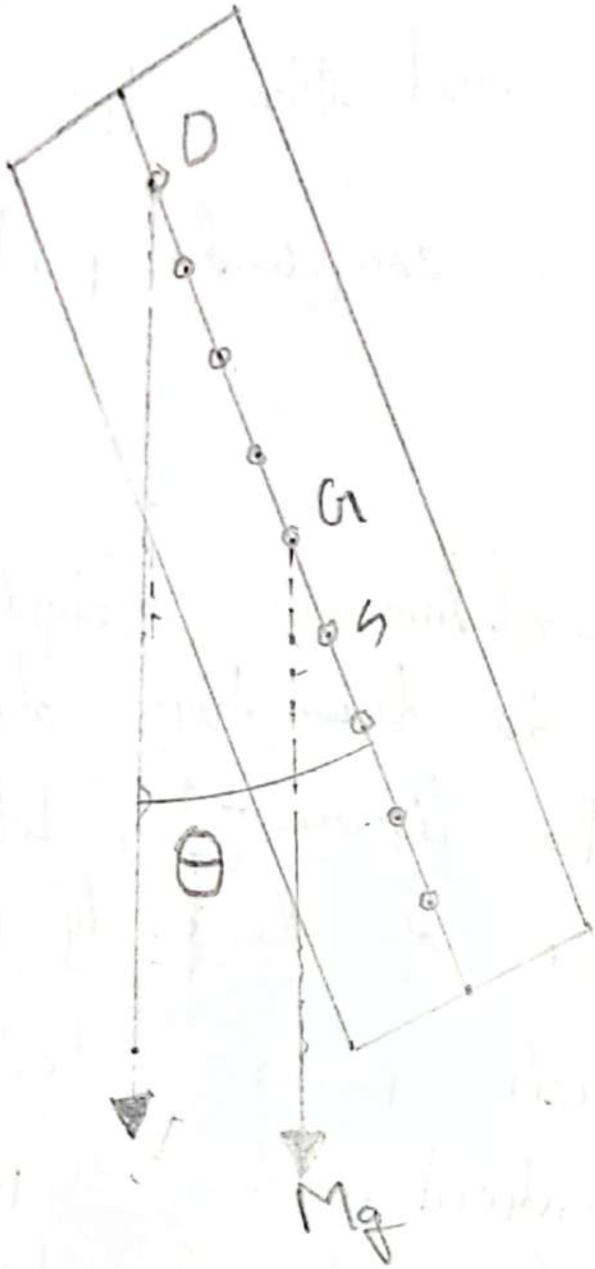


Figure-1: Compound Pendulum

Where $l = OG$.

If the angular acceleration produced in the body by this couple be $\frac{d\omega}{dt}$, the couple will ~~be~~ also be equal to $I\frac{d\omega}{dt}$, where I is the moment of inertia of the body about an axis through the point of Suspention and perpendicular to the plane of paper.

So that, $I\frac{d\omega}{dt} = -Mgl\sin\theta = -Mgl\theta$ .....(1)

or, $\frac{d\omega}{dt} = -\frac{Mgl\theta}{I} = -\mu\theta$ ......(2)

Thus, the angular acceleration $\left(\frac{d\omega}{dt}\right)$ of the body is proportional to its angular displacement (θ). The body, therefore, exicutes a S.H.M.

for ~~S.M~~. S.H.M. we know that

Acceleration $= -\omega^2 \times$ displacement ......(3)

From equation (2) and (3), we have

$\omega^2 = \frac{Mgl}{I}$ or, $\left(\frac{2\pi}{T}\right)^2 = \frac{Mgl}{I}$ therefore, $T = 2\pi\sqrt{\frac{I}{Mgl}}$ .....(4)

If k is the radius of ~~grav~~ gyration of the pendulum about an axis through G parallel ~~to the~~ to the axis of oscillation through O, from the Parallel Axes Theorem,

$I = M(k^2 + l^2)$

And so $T = 2\pi\sqrt{\frac{k^2+l^2}{gl}} = 2\pi\sqrt{\frac{\frac{k^2+l^2}{l}}{g}}$ .... (1)

Since the periodic time of ~~a~~ a simple pendulum is given by ~~T~~ $T = 2\pi\sqrt{\frac{L}{g}}$

The period of the rigid body (Compound pendulum) is the same as that of a ~~so~~ simple pendulum ~~Lenght~~ length $L = \frac{k^2+l^2}{l}$

This ~~Lenght~~ length L is known as the length of the simple equivalent pendulum. The expression for L can be written as a quadratic equation in (l). Thus from (2),

$\cancel{L^2 - lk + l}$ $L^2 - lL + k^2 = 0$ .... (3)

This give two values of l (l_1 and l_2) for which the body has equal times of vibration (time period). From the theory of quadratic equation,

$$l_1 + l_2 = L \quad \text{and} \quad l_1 \cdot l_2 = K^2$$

As the sum and product of two roots are positive, the two roots are both positive. This means that there are two positions of the center of suspension on the same side of center of gravity (C.G.) about which the periods (T) would be same.

Similarly there will be two more points of suspention on the other side of the C.G., about which the time periods (T) will agin be the same. Thus there are ~~all~~ altogether four points, two on either side of the C.G., about which the time periods of the pendulum are the same (T).

The distance between ~~twos~~ two such points, asymmetrically situated on either side of the C.G., will be the length (L) of the simple equivalent pendulum. If the length OG in figure 1, is l_1 and we measure the length

$GS = \frac{k^2}{l_1}$

along OG produced, the obviously $\frac{k^2}{l_1} = l_2$

or $OS = OG + GS = l_1 + l_2 = L$

The period of oscillation about either O or S is the same. The point S is called the center of oscillation. The points O and S are interchaneable i.e., when the body oscillates about O or S, the ~~th~~ time period is the same. If this period of oscillation is T, then from the expression, $T = 2\pi\sqrt{\frac{L}{g}}$

we get, $g = 4\pi^2 \frac{L}{T^2}$

By fingding L graphically and determining the value of the period. T, the acceleration due to gravity (g) at the place of the ~~exp~~ experiment can be determined.

Apperatus : A bar pendulum,

A small metal nail,

A spril level,

Stop watch

And a prism.

Data Sheet:

(A) Center of gravity of the compound pendulum = 50 cm

(B) Table for the **time period** T and the **distance** of the **point of suspension** from **end** A.

	Hole No From end A	Distance from end A l cm	Time for 10 oscillations t sec	Mean Time Period $T = t/10$ Sec
End A	1	5	16.27	1.627
	2	10	15.31	1.53
	3	...	14.5	1.45
	4	...	14.2	1.42
	5	...	14.9	1.49
	6	...	16.1	1.61
	7	...	19.5	1.95
	8	40	2.31	2.31
	9	45	27.15	2.715
End B	11	55	26.7	2.67
	12	60	22.8	2.28
	13	...	19.36	1.936
	14	...	16.2	1.62
	15	...	14.82	1.482
	16	...	14.3	1.43
	17	....	14.26	1.426
	18	90	15.1	1.51
	19	95	Calculation 16.17	1.617

19.11.24

Calculation:

Method of measuring length (L):
(From graph)

Length, **AD** = ...65.... cm.

Length, **BE** = ...65.... cm.

Mean length, **L = (AD+BE) / 2** =....65....cm = ...0.65....m

Corresponding time-period from the graph. (Time corresponding the **ABCDE** line)

T = ...1.627....sec.

$$g = 4\pi^2 \frac{L}{T^2} = \frac{4\times\pi^2\, 0.65}{1.627^2} = 9.69 \text{ m/sec}^2$$

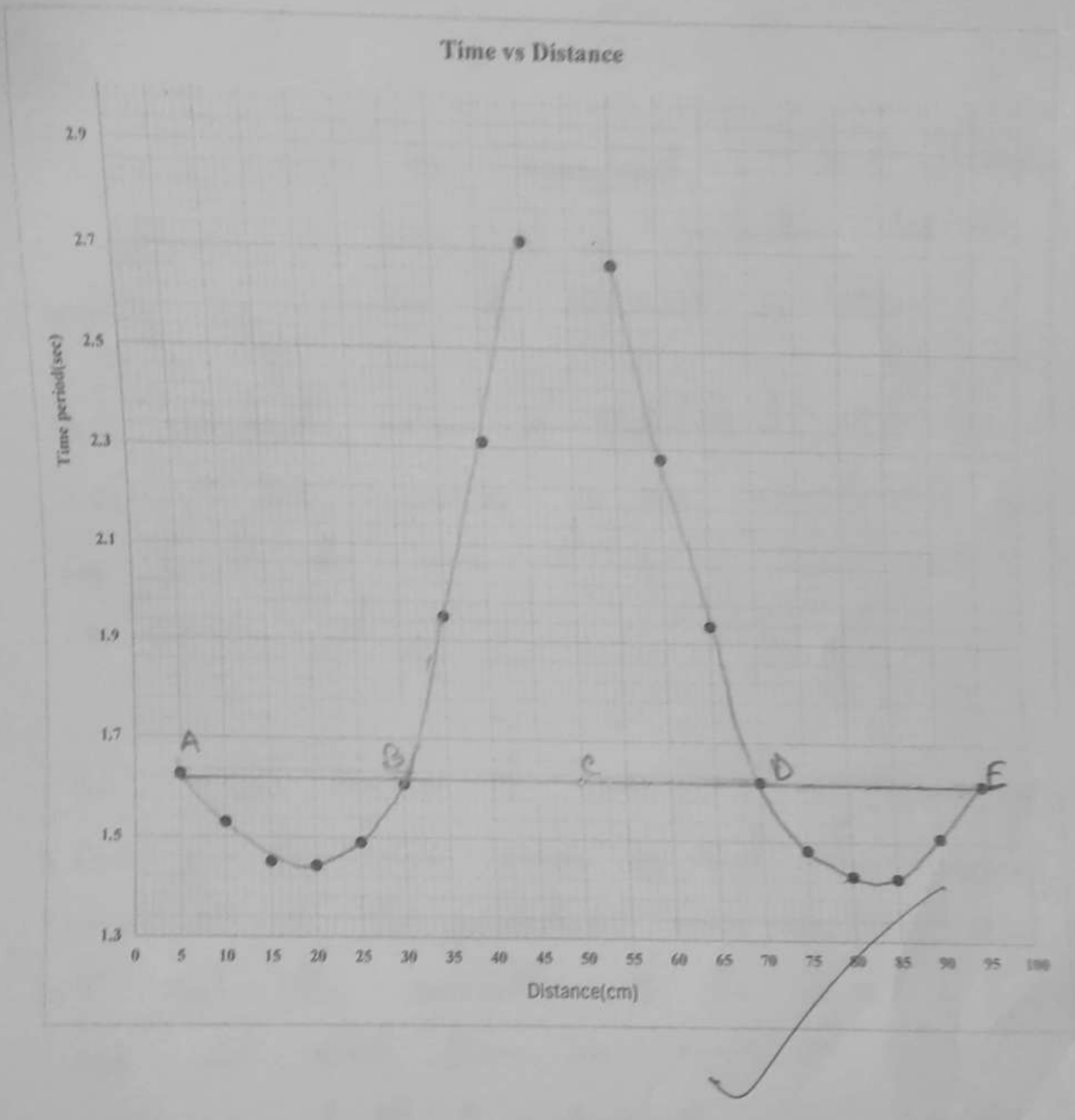
Standard value (Theoretical value) of **g = 9.81 ms^{-2}**

$$\text{Percentage Difference} = \frac{\text{Standard value} \sim \text{Experimental value}}{\text{Standard value}} \times 100\%$$

Percentage difference = ...1.2%....

Result:

Acceleration due to gravity, g = ...9.69 ms^{-2}....



Time vs Distance
Time period(sec)
2.9
2.7
2.5
2.3
2.1
1.9
1.7
1.5
1.3
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
100
Distance(cm)
A
B
C
D
E

Result :

Acceleration due to gravity, $g = 9.69\ ms^{-2}$

~~Ref~~

Discussions : In this experiment we have attempted to determine the value of g, acceleration due to gravity by means of compound pendulum.

The standard value of gravitational acceleration is $9.81\ ms^{-2}$ but according to our experiment's results we found the value of g is $9.69\ ms^{-2}$ with an error of 1.1%.

The major sources of error and possible solutions....

1. One of the main error is that the angular amplitude of the pendulum must be less than 5°. But the oscillation of the compound pendulum did not start from an angle of 5°.
2. Stopwatch should be observed properly and oscillation should be counted properly as both play a vital role in this experiment.
3. Internal mechanical friction might entrap the normal movement of the pendulum for the period might have changed.